

Basic Motor Parameters Adjustment

1. Parameter reset:

Reset Pr.00-02 = 10 (60 Hz) to the default value.

2. Select PM motor type:

Pr.05-33 = 1 (SPM) or 2 (IPM)

3. Motor nameplate parameter setting:

Parameter	Description
Pr.01-01	Rated frequency (Hz)
Pr.01-02	Rated voltage (V _{AC})
Pr.05-34	Rated current (A)
Pr.05-35	Rated power (kW)
Pr.05-36	Rated rotor speed (rpm)
Pr.05-37	Number of poles for the motor (poles)

4. PM parameter auto-tuning:

Set Pr.05-00 = 5 (Rolling auto-tuning for PM) or 13 (Static auto-tuning for PM) and press “RUN” key to finish motor auto-tuning, then you will get the following parameters:

Parameter	Description
Pr.05-39	Stator resistance for a permanent magnet motor (Ω)
Pr.05-40	Permanent magnet motor Ld (mH)
Pr.05-41	Permanent magnet motor Lq (mH)
Pr.05-43	Ke parameter of a permanent magnet motor (V _{phase . rms} / krpm) (When Pr.05-00 = 5, the Ke parameter is measured based on the actual motor rotation.) (When Pr.05-00 = 13, the Ke parameter is automatically calculated based on the motor power, current and rotor speed.)

If an auto-tuning error (AUE) occurs, refer to Section 14 “Fault Codes and Descriptions” for further treatment.

AUE Error (code)	Description
AUE (40)	Auto-tuning error
AUE1 (142)	Auto-tuning error 1 (No feedback current error)
AUE2 (143)	Auto-tuning error 2 (Motor phase loss error)

5. Set control mode

Control mode for the motor: Pr.00-11 = 2: PM SVC mode

6. Measure the initial magnetic pole angle of PM

Set Pr.10-53 PM initial rotor position detection method

0: Disable

1: Using I/F current command (Pr.10-31) to attract the rotor to zero degrees

2: High frequency injection

3: Pulse injection

(Set to 2 for IPM; set to 3 for SPM. If these settings cause problems, then set the parameter to 1.)